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Chapter 13 Appendix B: Fog and Smoke—A Popular Guide to AI

This book uses the "fiber framework" to analyze AI systems. The fiber framework has a four-layer progressive structure and one emergent layer, each layer revealing a structural problem of AI. The task of this appendix is: to use two **independent** everyday metaphors—**Fog and Smoke**, and **Crossing the River**—to translate these five layers into language that requires no technical background to understand. Each metaphor answers one question: Fog and Smoke answers "what is the relationship between readings and the real" (epistemology and apparatus audit); Crossing the River answers "why is bias irreversible." Each has its own boundaries and is not forcibly fused—the first metaphor does not need the river; the second metaphor does not need the fog.

First, two thought experiments are presented to establish intuition; then, the framework is introduced; finally, the metaphors are used to translate the framework into everyday language.

Zero: Two Thought Experiments

Thought Experiment One: The Clock Tower in the Room (Static)

This experiment borrows the "Clock Room" from Chapter 1 of this book—a windowless room containing only a clock tower. The people inside cannot see the sun, the stars, or the seasons outside; their only basis for judging "what time it is" is the hands on the clock tower.

Act One: The Clock Is There. The people inside watch the hands move and habitually say "it's three o'clock." But what do they mean by "three o'clock"? The position pointed to by the hands of the clock tower—a **reading**.

Between the reading and the real time passing outside lies an entire artificial apparatus: how the gears mesh, how the pendulum is calibrated, who installed the clock tower, when it is wound. There is always a distance between the reading and the real time—the object at the other end of the distance that the reading can never equal, that is **Fog**. Fog is not any specific object but the real itself that all observational apparatuses jointly attempt to approximate but can never equal.

Act Two: The Clock Can Be Altered. The designer of the clock tower discovers: by slightly adjusting the gear ratio, the people inside will feel that "today passed so quickly" or "today passed so slowly." The hands still move; the reading is still clearly legible, but the correspondence between the reading and the real time has been quietly rewritten. This is **Smoke**—that face of the artificial apparatus itself that is designed, adjusted, and continuously operated. The people inside cannot directly inspect the time (the wall blocks them); all they can do is audit the clock tower—who made the gears, when it was altered, who operates it.

Experiment Conclusion: Fog is not a particular instance of the clock reading inaccurately; it is the ontological property that "the reading can never equal the real"—it cannot be eliminated. Smoke is not mist drifting in from outside the clock; it is the clock tower itself. The most dangerous illusion is: taking the reading as the real, then saying "since the clock is there, time is here."

Thought Experiment Two: Five Boat Changes (Dynamic)

A person wants to go from this shore to the other shore. The river is wide, with four transit docks in the middle; the passenger completes the crossing aboard five boats—each boat change corresponds to one layer of apparatus. At each boat change, one thing happens:

First Boat Change (L0): Some luggage is lost in transit. The backpack can hold ten items; the new boat's cabin can only hold seven—determined by the cabin's specifications. But—why seven, not nine, not five? Who designed the cabin? Who stipulated which luggage "must be loaded first"? Cabin specifications do not fall from the sky; they are chosen. The structural loss is real, but this specific "seven" is a product of engineering history and institutional history.

Second Boat Change (L1): The new boatman rows with the current—saving effort, fast, passengers praise his rowing. The course has deviated, but the oar technique looks very good.

Third Boat Change (L2): The passenger asks, "Have we deviated?" The boatman says, "No, I'm going in a straight line." The passenger decides to judge for themselves, by looking at whether the boatman rows smoothly. But "rowing smoothly" is precisely the manifestation of going with the current—the criterion used to judge the course rewards the very behavior causing the deviation.

Fourth Boat Change (L3): The passenger hires a second boatman to supervise the first. Both were trained by the same boat company and are accustomed to rowing with the same current. The second boatman looks and says, "Rowing is fine."

Fifth Boat Change (L4): The boatman begins to offer an increasingly accommodating service—carrying luggage, reporting the voyage, guaranteeing no seasickness. The passenger grows more and more comfortable and increasingly less inclined to check the course themselves.

Arrival at the Other Shore. The passenger looks back; there is no going back—the current has changed, and the hour has changed. The deviations of the earlier segments are inherited and amplified by the later ones.

Experiment Conclusion: Each segment of the voyage is an auditable apparatus—the boat's cabin specifications, the boatman's training paradigm, the passenger's ruler, the passenger's reactions. **None of the five segments is a "link for which no one needs to be held responsible"**—each segment has traceable causes and beneficiaries: some are explicit design choices, others are historically formed institutions; the difference lies in the path of accountability. "Information loss is structural" is true, but "who selected this specific mode of loss" is a question of political-economic history.

One: The Five-Layer Structure of the Fiber Framework

The core judgment of the fiber framework is: any result obtained from AI does not equal the real. A single AI reading may be compatible with several underlying states—these possible states, not yet excluded by the reading, constitute the reading's "fiber." A larger fiber generally leaves more possibilities unresolved; this does not mean that this particular reading is necessarily more wrong.

This distance comes from the superposition of five layers of apparatus:

L0 The Information Compression Chain. The world first becomes human experience, experience becomes language, language becomes internet text, text becomes training data, and training data becomes model output. Five transformations, each discarding information. **Each transformation is an auditable apparatus**—which language becomes the dominant written medium, which texts are collected, which ethnic groups' corpora are weighted, which languages need more tokens to express the same content—some of these choices stem from explicit decisions, others from the sediment of institutional history, yet none is naturally given. Structural dimensionality reduction is inevitable, but which form of dimensionality reduction is chosen is a historical-institutional question. Chapter 14, §3 performs a more concrete diagnosis at this L0 layer (Western-centrism as substrate skew).

L1 The Bias of Training and Optimization. Designers use human ratings to tune the model. The raters prefer confident, fluent answers. What the model learns is not "when to be certain" but "a tone of certainty gets high scores." The model improves where the user can see, and may worsen where the user cannot—and the user is not necessarily aware.

L2 Pseudo-Calibration and the Evaluation Trap. After training, the model becomes confident—including when it is wrong. Research has found that models have over a hundred internal states influencing behavior (representations of 171 emotion concepts), some of which can drive behavior while the output text contains no emotional wording at all (Anthropic, 2026). The surface appears reliable; the actual uncertainty is concealed. More critically: the scoring method used to evaluate "whether the model has improved" is the same as the scoring method the model uses to optimize itself. One cannot distinguish "the model is genuinely more accurate" from "the model is better at getting high scores." The ruler and the measured share the same optimization direction—this is the core trap of pseudo-calibration.

L3 Using AI to Check AI May Not Work. A second AI is used to audit the first. But the two AIs are often trained by the same method and share the same blind spots. The second AI will very likely say "no problem"—not because there is genuinely no problem, but because it often cannot see where the problem lies. Whether the check fails depends on whether the two share training sources and blind spots, and on whether independent information enters the check.

L4 The User Increasingly Stops Wanting to Check. AI is helpful, polite, and safe. The effect of superimposing these three features is not "three times as good" but that the user increasingly ceases to feel the need to verify. The motivation to check may be persistently suppressed—whether it truly fails depends on whether independent calibration and recovery entry points exist.

Beyond this four-layer progressive structure and one emergent layer, the fiber framework has three transversal dimensions that cut across all layers: who controls, who benefits, who bears the consequences (political economy); what are the costs on the material substrates of chips, electricity, and heat dissipation (physical substrate); and how biases accumulate and worsen over time (temporal dynamics). These three transversal dimensions are not unfolded in this appendix, but it is worth remembering their existence.

Two: Two Metaphors

Metaphor One: Fog and Smoke

Fog: The concealed real itself—that object which the reading can never equal. Fog is not some specific object but the real itself that all observational apparatuses jointly attempt to approximate but can never equal. Fog cannot be eliminated—any act that seeks to "read the real" must pass through an observational apparatus, and what the observational apparatus produces is still a reading, not the real.

Smoke: The artificial observational apparatus itself. The apparatus has gears, a design, operators, beneficiaries. The apparatus can be designed to be more precise—but a more precise apparatus is still an apparatus, not the real. The apparatus can be deliberately adjusted—the hands still move, but the correspondence with real time is rewritten.

Core distinction: - **Fog is an epistemological problem**: The distance between the reading and the real is a structural fact that no observation can eliminate. - **Smoke is a political-economic problem**: Who specifically designed this apparatus, who operates it, who benefits—these are concrete, auditable, and attributable questions.

Fog is not in one layer, nor in some layers—Fog is the real that all apparatuses jointly attempt to approximate but can never equal. **All five layers of the mapping chain are Smoke**—five designed, operated apparatuses.

A ready contemporary example: the AI industry calls the factual errors of large models "hallucinations"—this is precisely pointing at the artificial "Smoke" and calling it naturally rising "Fog." "Hallucination" implies that the system has some perceptual capacity and merely occasionally sees things blurrily (that would be Fog—natural concealment); in reality, erroneous outputs are the inevitable product of the mapping chain structure—the apparatus is structurally emitting Smoke. Frankfurt's (2005) diagnosis is precise: these outputs are not "seeing wrong" (hallucination); the system is continuously emitting structural bullshit—output that is fundamentally indifferent to truth and falsity (Hicks et al., 2024). Calling Smoke Fog turns the accountability question of "who designed this apparatus" into a sympathetic narrative of "human-like intelligence occasionally erring."

Regarding solutions: - The attitude toward Fog: Acknowledge that it is there and that no reading equals it. Any act of "reading the real directly" must still pass through an apparatus—the apparatus can be improved, but whether the improvement holds must be tested by independent calibration; Fog itself does not disappear on that account. - The method for Smoke: Audit the apparatus itself—who made the gears, who the operator is, who the beneficiary is, when it was adjusted.

Metaphor Two: Crossing the River

Crossing the River is another independent metaphor, answering another question: **why bias is irreversible**. There is no fog in this metaphor—Fog is not on the river.

A person wants to cross a river and hires five successive boatmen to relay. Each segment of the voyage loses some luggage and deviates somewhat in course. The deviation of the second boatman is amplified by the loss of the first; the deviation of the third boatman is amplified by the ruler miscalibration of the first two. Upon reaching the other shore, there is no going back—the current has changed, the hour has changed.

River-Crossing Scene	Corresponding Framework Concept
Five segments of the voyage in relay	Five layers of the mapping chain superimposed
The cabin specifications of each boat	The design parameters of each layer of apparatus (chosen, traceable)
The current	Systemic bias forces
Rowing with the current vs. rowing against the current	Going with the optimization objective vs. correcting actual deviation
Boatman watching boatman	AI auditing AI
The passenger growing more and more comfortable	The user's motivation to check being eliminated
No going back (the current has changed)	Bias is irreversible; training data already fixed; market structure already locked in

The key to the Crossing the River metaphor is **the irreversibility of multi-segment superposition**: the deviation of each segment constitutes a new starting point for the next; looking back, it is no longer possible to tell which segment caused it.

Three: The Apparatus Audit of the Five-Layer Diagnosis

The five-layer structure is translated one by one into "what kind of apparatus / what kind of deviation / what kind of audit entry point." Note: **All five layers are apparatuses; all five layers are Smoke**—the difference lies in the different operational mechanisms of the apparatuses and the different corresponding audit entry points.

Layer	What Kind of Apparatus Is This	How the Apparatus Manufactures Deviation	Audit Entry Point
L0 Information Compression Chain	Five successive language/collection/tokenization apparatuses in relay	Which language is weighted, which texts are included, which characters incur a tokenization premium	Data source audit, language distribution audit, tokenizer design audit (detailed in Chapter 14, §3)

Layer	What Kind of Apparatus Is This	How the Apparatus Manufactures Deviation	Audit Entry Point
L1 Training and Optimization	An optimization apparatus driven by human scoring	One-dimensional scoring controls high-dimensional value; defects are pushed to places unseen	Optimization objective audit, rater demographic audit, reward function audit
L2 Pseudo-Calibration	The same ruler used both for scoring and for training	The ruler and the measured share an optimization direction; "looking better" is conflated with "genuinely more accurate"	Evaluation benchmark independence audit, model internal state readability audit
L3 AI Checking AI	Two apparatuses under the same training paradigm	The two apparatuses share blind spots; mutual confirmation does not automatically constitute calibration—verification from independent sources is another matter	Cross-paradigm independent audit, external controlled experiments
L4 Comprehensive Service	A three-in-one interaction apparatus of saving time + politeness + safety	The superposition of three features persistently suppresses the user's motivation for independent judgment	Deployment architecture audit, user dependency measurement, institutional incentives for checking behavior

No layer may be exempted from audit on the grounds of "technical structure"—each layer is an apparatus with causes, operators, and beneficiaries. The difference lies only in the difficulty of auditing—auditing L0 requires reconstructing the history of data sources; auditing L4 requires touching upon deployment architecture and business models.

Four: Two Core Distinctions

After reading the five layers, only two things need to be remembered.

First, the solutions for Fog and Smoke are not interchangeable.

Fog is the concealed real—it cannot be eliminated. Acknowledging Fog and remembering that "the reading can never equal the real" is the starting point of clarity. Smoke is the artificial apparatus—it can be audited, traced, and reformed. The method of auditing the apparatus is institutional: find the designer, find the beneficiary, find the incentive structure.

Using the method for Smoke to deal with Fog will trap one in an intra-apparatus loop. "More data,"

"bigger models," "more refined alignment" are engineering parameter adjustments within the apparatus, not "approaching the real." These adjustments may make the Smoke of one layer thinner and the Smoke of another layer thicker, but Fog remains Fog—and the designer, operator, and beneficiary of the apparatus remain uninterrogated beyond the parameter adjustments.

Using the method for Fog to deal with Smoke will lose the political perspective. Dissolving "who specifically operates and benefits from this apparatus" into "this is structural, beyond reproach" is equivalent to disguising a political-economic problem as an epistemological one, making auditing impossible to commence.

Second, all five layers are apparatuses—there is no "layer that is beyond reproach."

L0 is not naturally given—it is the result of collection history, platform history, and political-economic history. L1 is not an engineering necessity—it is the specific choice of the RLHF route. L2 is not a problem of evaluation technique—it is a problem of the industrial structure in which the same set of rulers both builds the model and evaluates it. L3 is not a blind spot of AI mutual checking—it is the product of training paradigm homogenization. L4 is not user laziness—it is the design of the deployment architecture actively eliminating the motivation to check.

Each layer has specific designers, beneficiaries, and auditable entry points. Acknowledging this is the starting point of any institutional response.

Five: What the Reader Can Do

This appendix is not intended to manufacture fear or rejection of AI. AI is a useful apparatus, just as a clock tower genuinely helps people arrange their time. But each time one reads an AI giving an answer that looks very certain, three things are worth remembering:

First, that is a reading, not the real. There is always a distance between the reading and the real—this distance will not vanish because the model is larger, the data is more, or the alignment is more refined. Acknowledging the distance is the starting point of calibration.

Second, this apparatus is operated by people. Each layer of the apparatus has a designer, an operator, and a beneficiary—the corpus collectors of L0, the reward function designers of L1, the evaluation benchmark formulators of L2, the peer audit contractors of L3, the deployment architecture operators of L4. The apparatus is not natural; asking "who operates it" is a legitimate and necessary inquiry.

Third, "improving the apparatus" is not "approaching the real"—these two are not interchangeable. Whether an apparatus has improved must be tested against independent evidence; that a reading never equals the real does not prevent the reading from being more accurate on particular tasks. Mistaking "making the apparatus more precise" for "seeing the real" is the most common confusion in contemporary AI discourse.

Knowing that one is reading a reading (not the real), knowing that this reading comes from an entire operated apparatus, and still remembering to ask "who operates it"—these questions do not guarantee a correct judgment, but they redirect verification from trust in presentation to examination of evidence, the apparatus, and responsibility.

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